Domain-Adaptive Selective State-Space Learning for Cross-Subject Motor-Imagery EEG Decoding

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Abstract

Cross-subject motor-imagery electroencephalography (MI-EEG) decoding is hindered by low signal-to-noise ratios, scarce labeled data, and strong distribution shifts between subjects and sessions. This study proposes SSM-DAN, a domain-adaptive selective state-space network that couples a multi-scale convolutional front-end with bidirectional selective state-space blocks to capture both local and long-range temporal patterns, and reduces subject shift through hierarchical alignment, including a learned source–target normalization layer and label-free test-time adaptation. The framework was evaluated on three public MI-EEG datasets using leave-one-subject-out validation with chronologically separated training, adaptation, and test sessions. On BCI Competition IV-2a, it reached a mean accuracy of 76.12%, significantly higher than its source-only variant (64.35%; $p = 0.0039$) and its variant without test-time adaptation (72.15%; $p = 0.0039$). It also achieved 84.09% on BCI Competition IV-2b and 84.33% on Zhou2016, where it outperformed a matched TCFormer baseline on all four target subjects. Ablation studies showed that the contribution of each adaptation component depends on the dataset. These results indicate that combining selective state-space modeling with learned, hierarchical alignment improves cross-subject MI decoding without target labels.

Keywords: motor imagery, electroencephalography, brain–computer interface, selective state-space model, Mamba, domain adaptation, test-time adaptation

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1. Introduction

Motor-imagery brain–computer interfaces (MI-BCIs) decode movement intentions from non-invasive electroencephalography (EEG) for rehabilitation, assistive communication, and human–machine control. Their principal signatures, event-related desynchronization and synchronization, are weak, non-stationary, and distributed across time and electrodes [1], and they vary substantially between subjects and recording sessions. A model trained on existing users therefore often degrades on a new user, while collecting labeled calibration trials for every user is time-consuming. Cross-subject decoding must learn discriminative spatial–temporal structure from limited EEG and control inter-subject shift without target labels.

Classical pipelines such as common spatial pattern (CSP) and filter-bank CSP (FBCSP) rely on manually designed stages and subject-specific covariance structure that transfers poorly across users [2, 3]. End-to-end ConvNets and EEGNet learn temporal and electrode filters directly from EEG [4, 5] but relate distant time points only through depth, large kernels, or dilation. Hybrid decoders such as EEG Conformer, ATCNet, and TCFormer add self-attention and dilated temporal convolutional networks (TCNs) for global context [6, 7, 8], at a quadratic sequence cost [9] and with a high data demand in small BCI datasets. Selective state-space models provide linear-complexity sequence modeling [10, 11], and recent MI decoders combine convolution with Mamba [12, 13]; however, these decoders mainly study the architecture, use forward-only scans although complete offline trials are available, and do not address subject shift.

Domain generalization learns subject-invariant features from source subjects only [14, 15, 16], whereas unsupervised domain adaptation (UDA) also exploits unlabeled target EEG through Riemannian or Euclidean alignment [17, 18], MK-MMD [19], or adversarial learning with gradient reversal [20, 21]. HADANet combines the latter two for MI-EEG [22], prototype-based methods exploit class structure [23], and AutoDIAL aligns domains inside normalization layers by learning how to mix source and target statistics [24]. Test-time adaptation further reduces residual shift at inference through information maximization [25] or normalization-only updates [26]. Alignment with a fixed strength, however, risks negative transfer when the shift is small, alignment of pooled trial features alone can overlook temporal mismatch, and test-time adaptation accesses evaluation-session signals, which must be disclosed and kept separate from labels.

To our knowledge, selective state-space MI decoders have not been coupled with target adaptation, and existing EEG adaptation methods neither learn how strongly to align nor align temporal representations. We address these gaps with SSM-DAN, a selective state-space model with domain adaptation that couples a bidirectional selective state-space encoder with hierarchical, label-free adaptation, evaluated under a chronological leave-one-subject-out (LOSO) protocol in which target labels remain hidden throughout preprocessing, training, checkpoint selection, and adaptation. The contributions are:

- an end-to-end decoder that combines multi-scale convolution, alternating forward/backward selective state-space blocks, and local–global TCN fusion;
- a hierarchical adaptation strategy that combines Riemannian alignment, an AutoDIAL normalization layer, adversarial and MK-MMD alignment, and constrained test-time adaptation within a chronological, leakage-controlled LOSO protocol; and
- an evaluation on three public MI-EEG datasets with subject-wise accuracy, Cohen's κ, statistical tests, efficiency analysis, and component ablations.

2. Method

SSM-DAN has two parts (Fig. 1). An EEG encoder extracts local spectral–spatial features with multi-scale convolutions, models long-range temporal dynamics with bidirectional selective state-space blocks, and fuses both through a temporal convolutional head (Sections 2.1–2.3). A hierarchical adaptation scheme then reduces the shift between subjects at three levels: the input signal is aligned by Riemannian whitening, the learned features by an AutoDIAL normalization layer together with adversarial and discrepancy losses, and the predictions by test-time adaptation (Sections 2.4–2.7). The model is trained on labeled trials of the other (source) subjects and unlabeled trials of the new (target) subject (Stage 1), adapted to the unlabeled trials on which it will be evaluated (Stage 2), and then used for prediction (Stage 3).

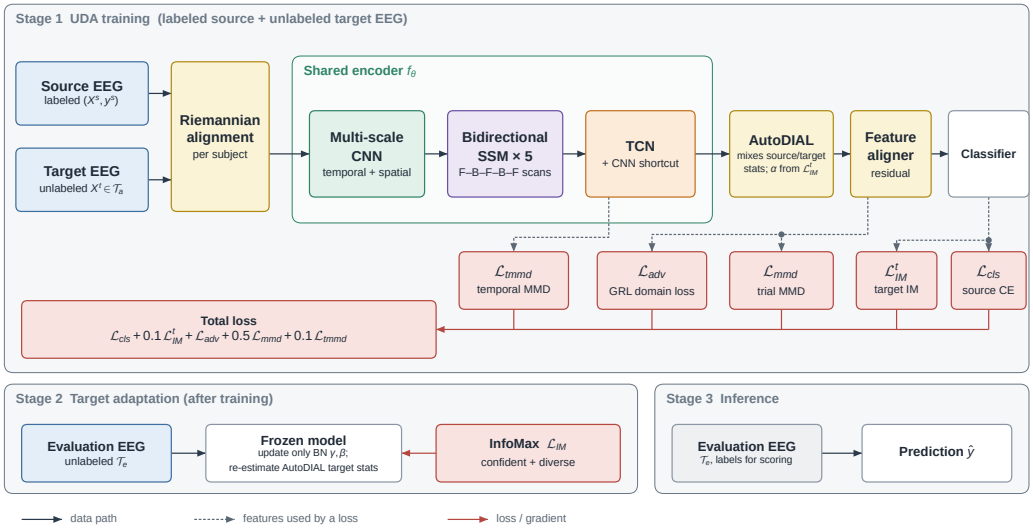


Figure 1: Overview of SSM-DAN: training with labeled source and unlabeled target EEG (Stage 1), label-free test-time adaptation on the unlabeled evaluation sessions $\mathcal{T}_e$ (Stage 2), and prediction (Stage 3).

2.1. Multi-scale convolutional encoder

Each trial is an EEG matrix $X \in \mathbb{R}^{C\times T}$ with C electrodes and T time samples. Following TCFormer [8], three temporal convolutions with kernels of 20, 32, and 64 samples capture rhythms at different time scales and produce 32 feature maps each (Fig. 2). A depthwise convolution across all electrodes then learns two spatial filters per map, followed by average pooling and dropout. A grouped pointwise convolution reduces the 192 maps to 48 channels, 16 per temporal scale, and a grouped temporal convolution refines them. A residual channel-group attention weights the three scales by one learned coefficient each, and a second pooling stage yields the local feature sequence $H_c \in \mathbb{R}^{48\times L}$, where L is the number of pooled time steps.

2.2. Bidirectional selective state-space encoder

To capture dependencies across the whole trial, H_c is mixed across channels and passed through five selective state-space blocks [10, 11]. Each block normalizes its input and produces, through gated projections and a short

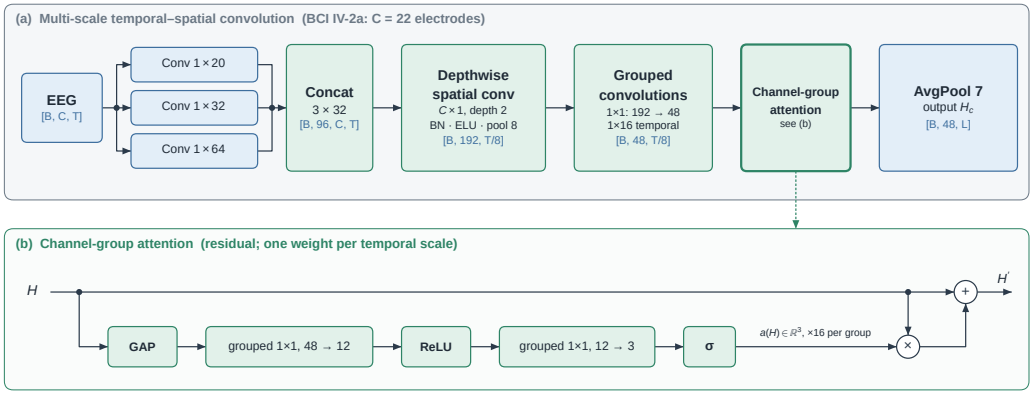


Figure 2: Multi-scale temporal–spatial front-end. (a) Three temporal branches, depthwise electrode filtering, grouped refinement, and channel-group attention produce H_c. (b) Channel-group attention: grouped 1×1 squeeze–excitation yields one sigmoid weight per temporal scale, which rescales that scale's 16 channels and is added residually.

causal convolution, a sequence u_t that drives a diagonal recurrence:

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t \odot u_t, \tag{1}$$

$$z_t = C_t^\top h_t + d \odot u_t, \tag{2}$$

where h_t is the hidden state, $A = -\exp(A_{\log})$ is a learned negative decay that keeps the recurrence stable, and the step size Δ_t and the projections B_t and C_t are computed from the input. This input dependence lets each block decide which time steps to retain, and its cost grows linearly with L. Because complete trials are available offline, the blocks alternate between forward and backward scanning, a backward block reversing the time axis before and after its recurrence (Fig. 3). This gives bidirectional context without duplicating every block. The state dimension is eight, with dropout 0.4 and depth-dependent stochastic depth; the recurrence is implemented in plain PyTorch rather than with the hardware-specific Mamba kernel.

2.3. Local–global fusion and classification

The state-space output is reduced to 16 channels and concatenated with the 48-channel local sequence H_c, so that the head sees both local and global evidence. Two residual grouped temporal convolutional network (TCN) blocks with kernel length four and dilations one and two refine the resulting 64-channel sequence. The average of its last time step and its temporal

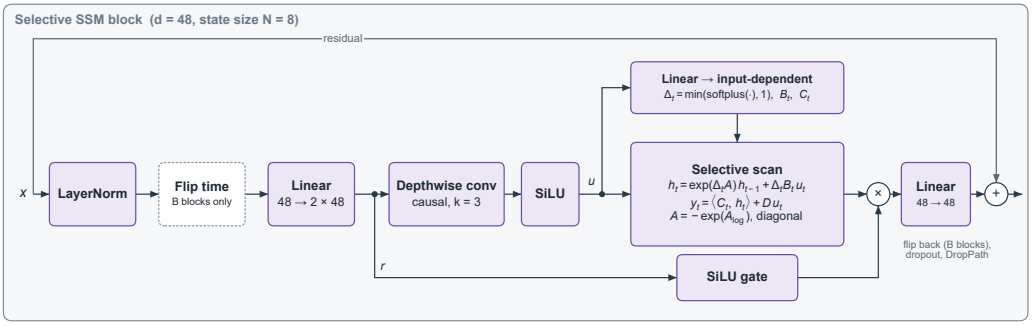


Figure 3: Selective SSM block. Backward blocks reverse and restore the temporal axis around the mixer; forward blocks omit these operations.

mean forms the trial representation $q \in \mathbb{R}^{64}$, made of four 16-channel groups (three convolutional scales and the state-space branch). A norm-constrained grouped classifier produces one set of class scores per group, combined with learned softmax weights.

2.4. Riemannian signal alignment

Different subjects produce EEG covariance structures that are shifted relative to one another. In Riemannian alignment (RA), each subject's trials are therefore whitened by the inverse square root of that subject's mean covariance $\bar{R}$, computed as the affine-invariant Riemannian mean of the OAS-shrunk trial covariances [17, 18]:

$$\widetilde{X} = \bar{R}^{-1/2} X. \tag{3}$$

To avoid leakage, $\bar{R}$ is estimated only from data available before evaluation. For a source subject, it is computed from the training sessions and applied to the validation sessions. For the target subject, it is computed from the earlier, unlabeled sessions $\mathcal{T}_a$ and applied unchanged to the later sessions $\mathcal{T}_e$ used for evaluation.

2.5. Automatic domain alignment layer

Riemannian whitening aligns second-order input statistics but cannot correct shifts that arise inside the network. Following AutoDIAL [24], the trial representation q is therefore normalized by an alignment layer that lets the network decide how strongly target features should be standardized with their own statistics. Source features use source mean and variance (μ_s, σ_s^2);

target features use the moments of a mixture of the source and target (μ_t, σ_t^2) distributions,

$$\mu_\alpha = \alpha\mu_t + (1-\alpha)\mu_s, \tag{4}$$
$$\sigma_\alpha^2 = \alpha\sigma_t^2 + (1-\alpha)\sigma_s^2 + \alpha(1-\alpha)(\mu_t - \mu_s)^2, \tag{5}$$

so that

$$\hat{q}^s = \gamma \odot \frac{q^s - \mu_s}{\sqrt{\sigma_s^2 + \epsilon}} + \beta, \qquad \hat{q}^t = \gamma \odot \frac{q^t - \mu_\alpha}{\sqrt{\sigma_\alpha^2 + \epsilon}} + \beta, \tag{6}$$

with shared affine parameters γ and β. The mixing weight $\alpha \in (0, 1)$, initialized to 0.5, is learned separately for each of the four feature groups: $\alpha \to 0$ normalizes target features with source statistics, whereas $\alpha \to 1$ normalizes them entirely with their own. During training the moments are computed per mini-batch and tracked as running averages (momentum 0.1); at inference the running moments are mixed with the learned α, and source validation trials always use source statistics.

2.6. Training objective

A residual $64 \to 128 \to 64$ aligner maps the normalized representation to the space used by the classifier and the alignment losses. The source cross-entropy $\mathcal{L}_{\text{cls}}$ is the only labeled term. A domain discriminator trained through gradient reversal [20] provides $\mathcal{L}_{\text{adv}}$, whose weight rises smoothly from zero to one during training, and a five-kernel MK-MMD [19] reduces the distance between source and target features, both for trial representations ($\mathcal{L}_{\text{mmd}}$) and for time-averaged TCN sequences ($\mathcal{L}_{\text{tmmd}}$). Finally, the target predictions are encouraged to be confident and class-balanced by the information-maximization loss [25]

$$\mathcal{L}_{\text{IM}} = \frac{1}{N}\sum_{j=1}^{N} H(p_j) - H\left(\frac{1}{N}\sum_{j=1}^{N} p_j\right), \tag{7}$$

where p_j is the predicted class distribution of the j-th of N unlabeled trials and $H(\cdot)$ is the entropy; the first term rewards confident predictions and the second prevents all trials from being assigned to one class. Applied to a target mini-batch, it gives $\mathcal{L}_{\text{IM}}^t$, and the training objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 0.1\mathcal{L}_{\text{IM}}^t + \mathcal{L}_{\text{adv}} + 0.5\mathcal{L}_{\text{mmd}} + 0.1\mathcal{L}_{\text{tmmd}}. \tag{8}$$

The target trials in this stage come only from $\mathcal{T}_a$. The alignment losses see the AutoDIAL output with α held fixed, since they could otherwise be reduced trivially by setting α to one; α is therefore learned only through $\mathcal{L}_{\mathrm{IM}}^{t}$, toward the normalization that yields confident yet class-diverse target predictions.

2.7. Test-time adaptation

After training, the checkpoint with the highest source-validation accuracy is adapted to the evaluation trials $\mathcal{T}_e$ without their labels by information-maximization test-time adaptation (IM-TTA). All parameters except the BatchNorm scales and shifts are frozen, and Eq. (7) is minimized over $\mathcal{T}_e$ for five Adam passes with learning rate 10^{-4}; restricting the update to normalization parameters limits the risk of overfitting a small target set [26]. Because this changes the backbone features, the AutoDIAL target moments (μ_t, σ_t^2) are then re-estimated in one pass over the same trials, while α and the source moments remain fixed. The labels of $\mathcal{T}_e$ are used only to score the final predictions.

3. Experimental Setup

3.1. Datasets

Three public MI-EEG datasets with different numbers of electrodes, classes, subjects, and sessions are used (Table 1). All are accessed through MOABB/Braindecode, which provides a common representation of subjects, sessions, runs, and event windows [27].

Table 1: Summary of the MI-EEG datasets (EOG channels excluded).

Dataset	Subjects	Sex (M/F)	Channels	Sessions	Classes
BCI IV-2a	9	n.r.	22	2	left hand, right hand, feet, tongue
BCI IV-2b	9	n.r.[a]	3	5	left hand, right hand
Zhou2016	4	1/3	14	3	left hand, right hand, feet

n.r.: not reported in the dataset description. [a]The nine participants were drawn from a study of ten (6/4, 24.7 $\pm$ 3.3 years) [28]; the split of the nine released participants is not reported.

3.1.1. BCI Competition IV-2a

The dataset contains nine participants recorded in two sessions on different days; each session has 288 trials distributed equally over imagined left-hand, right-hand, both-feet, and tongue movements. EEG was recorded from 22 electrodes (three additional EOG channels are not used) at 250 Hz with hardware band-pass filtering at 0.5–100 Hz and a 50-Hz notch [29]. The two sessions are denoted session 1 and session 2. Participant sex and age are not reported.

3.1.2. BCI Competition IV-2b

The dataset contains two-class left- versus right-hand MI from nine right-handed participants with normal or corrected-to-normal vision, drawn from a study of ten participants (six male, four female; age 24.7 ± 3.3 years) [28]; the sex of the nine released participants is not reported individually. Three bipolar derivations around C3, Cz, and C4 (three EOG channels are not used) were sampled at 250 Hz. Sessions 1–3 form the original calibration portion and sessions 4–5 the evaluation portion [29], a chronological separation that is kept here. The dataset tests whether the method remains useful with only three electrodes.

3.1.3. Zhou2016

The dataset contains four healthy participants (one male and three female, aged 22–28 years; sex reported only at group level), each recorded in three sessions separated by days to months [30]. Each session has two runs of 75 trials (25 per class: left hand, right hand, and feet), recorded from 14 electrodes at 250 Hz. Each trial contains a 5-s cue period, during which the imagery is performed, followed by a 4-s rest.

3.2. Chronological LOSO protocol with test-time adaptation

One subject is held out in each LOSO fold. All remaining subjects are sources; the held-out participant is the target. Table 2 specifies the chronological session split. Earlier target sessions are used without labels during Stage 1 UDA. The later evaluation sessions remain unavailable during model training, preprocessing-statistic fitting, and checkpoint selection; at inference time, their EEG signals are passed without labels to IM-TTA and AutoDIAL target-statistic re-estimation, after which the adapted model is scored using the withheld labels. This is the primary protocol for headline comparisons.

Table 2: Chronological session allocation in each LOSO fold. Evaluation-session EEG is available without labels only to IM-TTA and AutoDIAL target-statistic re-estimation at inference time; its labels are used afterward solely for scoring.

Dataset	Source subjects (labeled)		Target subject	
	Training	Validation	Stage 1 UDA (unlabeled)	IM-TTA + evaluation (labels withheld)
BCI IV-2a	session 1	session 2	session 1	session 2
BCI IV-2b	sessions 1–3	sessions 4–5	sessions 1–3	sessions 4–5
Zhou2016	sessions 1–2	session 3	sessions 1–2	session 3

This split yields nine folds for BCI IV-2a and BCI IV-2b and four for Zhou2016, with identical split rules and hyperparameters in every fold. The checkpoint with the highest source-validation accuracy is selected before IM-TTA, and neither evaluation-session labels nor evaluation metrics participate in adaptation or checkpoint selection. Stage 1 adaptation (AutoDIAL, target information maximization, and the adversarial and MK-MMD losses) uses only the earlier target sessions, whereas Stage 2 IM-TTA and AutoDIAL re-estimation use the later evaluation sessions immediately before scoring. For Zhou2016, an all-session target score is additionally reported as an auxiliary diagnostic only.

3.3. Preprocessing and augmentation

Only EEG channels are retained. Signals are resampled to 250 Hz and converted from volts to microvolts. The BCI Competition datasets receive no additional digital band-pass filtering: BCI IV-2a uses the complete MOABB event window, while BCI IV-2b removes 0.5 s from its end. For Zhou2016, all 14 EEG channels are band-pass filtered at 8–30 Hz and the interval from 0.5 to 5.0 s after cue onset is retained (4.5 s, or 1125 samples). This setting suppresses the initial cue transient and focuses both feature extraction and covariance estimation on the motor-imagery μ/β range.

Riemannian whitening, when active, follows the session-isolated fitting rule in Section 2.4. Z-score statistics are fitted using the source-training partition and then frozen. Following TCFormer [8], the segmentation and reconstruction (S&R) augmentation introduced by Lotte [31] is applied to labeled source mini-batches of BCI IV-2a and IV-2b: each trial is divided into

eight consecutive, non-overlapping segments, and synthetic trials are formed by concatenating, for each segment position, the segment of a randomly chosen trial of the same class, which doubles the mini-batch. It neither reads target labels nor mixes source and target trials. Zhou2016 is run without this augmentation to preserve its intended dataset-specific setting.

3.4. Implementation details

All experiments are executed in Kaggle notebooks using one NVIDIA Tesla T4 GPU with 16 GB memory. BCI IV-2a and BCI IV-2b use a mini-batch size of 48 and a maximum of 125 epochs. Zhou2016 uses a mini-batch size of 32 and 100 epochs. The BCI Competition runs use Adam with an initial learning rate of 9×10^{-4}, $\beta_1 = 0.5$, and weight decay 10^{-3}; the Zhou2016 setting uses an initial learning rate of 10^{-3} and weight decay 10^{-4}. A linear warm-up over the first three LOSO epochs is followed by cosine decay. The random seed (0), data split, and initialization procedure are fixed and shared by all methods and ablation variants, so that every reported comparison differs only in the evaluated component.

3.5. Comparison methods

Comparison methods are grouped by the target data they may access, and each group is compared with the SSM-DAN variant that has the same access. Source-only decoders, namely EEG Conformer [6], ATCNet [7], TC-Former [8], EEGEncoder [32], and SATrans-Net [33], are trained on labeled source data only. Unsupervised domain adaptation (UDA) methods, namely CLUDA [34], RMETNet [35], and MSD-DDA [36], additionally use the unlabeled earlier target sessions during training, which is the target access of Stage 1 of SSM-DAN, and are evaluated without test-time adaptation; their matched reference is SSM-DAN (UDA only), i.e., SSM-DAN without test-time adaptation. Test-time adaptation methods, namely T-TIME [37] and BFT [38], train on source data only and adapt to the unlabeled evaluation trials at inference; their matched reference is SSM-DAN (TTA only), i.e., SSM-DAN without domain adaptation, which removes every Stage 1 adaptation component while retaining Riemannian alignment and IM-TTA. All methods follow the session split in Table 2, use the same preprocessing, and select the checkpoint with the highest source-validation accuracy.

3.6. Evaluation measures and reporting

Accuracy and Cohen's κ ($\times 100$) are reported per target subject and as mean $\pm$ standard deviation across LOSO folds. Paired accuracy differences use a two-sided Wilcoxon signed-rank test with exact sign-permutation p-values. Efficiency is summarized by trainable parameters, mean Stage 1 training time per fold on a Tesla T4, and batch-one CPU response time averaged over 500 passes after 100 warm-up passes; IM-TTA, preprocessing, and data transfer are excluded from these timings.

4. Results

Tables 3–5 report subject-wise chronological LOSO results. Standard deviations describe variation across target subjects, not repeated seeds.

4.1. BCI Competition IV-2a

Table 3 reports the four-class chronological-session LOSO results for the nine BCI IV-2a target subjects. SSM-DAN applies label-free IM-TTA to each target subject's session 2 EEG immediately before evaluation.

SSM-DAN achieved $76.12 \pm 11.96\%$ accuracy and 68.20 ± 15.90 Cohen's κ, outperforming its source-only variant ($64.35 \pm 12.77\%$) and its variant without test-time adaptation ($72.15 \pm 13.58\%$). Both paired accuracy differences were significant ($p = 0.0039$ for each comparison); IM-TTA improved all nine target folds. SSM-DAN also exceeded the matched EEG Conformer ($55.25 \pm 13.39\%$; $\kappa = 40.30 \pm 17.90$), ATCNet ($57.02 \pm 12.74\%$; $\kappa = 42.70 \pm 17.00$), TCFormer ($60.80 \pm 11.49\%$; $\kappa = 47.70 \pm 15.30$), EEGEncoder ($58.06 \pm 12.55\%$; $\kappa = 44.10 \pm 16.70$), and SATrans-Net ($62.00 \pm 10.34\%$; $\kappa = 49.30 \pm 13.80$) baselines on all nine targets ($p = 0.0039$ for each). The UDA baselines CLUDA, RMETNet, and MSD-DDA use the unlabeled target session 1 during training and no test-time adaptation, so they are compared with SSM-DAN (UDA only; $72.15 \pm 13.58\%$). The test-time adaptation baselines use no target data during training and are compared with SSM-DAN (TTA only), which reached $71.37 \pm 11.78\%$ ($\kappa = 61.80 \pm 15.70$): T-TIME reached $66.24 \pm 9.34\%$ ($\kappa = 55.00 \pm 12.50$) and BFT $63.08 \pm 9.77\%$ ($\kappa = 50.80 \pm 13.00$), 5.13 and 8.29 percentage points lower on average. The reference exceeded T-TIME on seven of the nine targets ($p = 0.0742$) and BFT on all nine ($p = 0.0039$). As an auxiliary diagnostic only, the variant without test-time adaptation obtained $73.32 \pm 11.94\%$ accuracy and 64.40 ± 15.90 Cohen's κ when target sessions 1 and 2 were pooled.

Table 3: Per-subject LOSO accuracy (%) and Cohen's κ multiplied by 100 on BCI IV-2a (four classes). p: Wilcoxon signed-rank test of accuracy against SSM-DAN for source-only methods and SSM-DAN variants, against SSM-DAN (UDA only) for unsupervised domain adaptation methods, and against SSM-DAN (TTA only) for test-time adaptation methods (Section 3.5). –: not yet available.

Method	Metric	Target subject									Mean ± SD	p
		S1	S2	S3	S4	S5	S6	S7	S8	S9		
Source-only training												
Conformer [6]	Acc.	58.68	43.06	78.82	43.06	44.44	41.67	72.92	65.28	49.31	55.25 ± 13.39	0.0039
	κ	44.91	24.07	71.76	24.07	25.93	22.22	63.89	53.70	32.41	40.30 ± 17.90	
ATCNet [7]	Acc.	69.44	40.97	76.74	50.35	43.40	42.01	61.81	70.49	57.99	57.02 ± 12.74	0.0039
	κ	59.26	21.30	68.98	33.80	24.54	22.69	49.07	60.65	43.98	42.70 ± 17.00	
TCFormer [8]	Acc.	65.28	49.65	79.86	54.86	61.81	38.89	71.88	66.67	58.33	60.80 ± 11.49	0.0039
	κ	53.70	32.87	73.15	39.81	49.07	18.52	62.50	55.56	44.44	47.70 ± 15.30	
EEGEncoder [32]	Acc.	64.24	49.31	78.82	46.18	54.86	40.97	74.31	65.97	47.92	58.06 ± 12.55	0.0039
	κ	52.31	32.41	71.76	28.24	39.81	21.30	65.74	54.63	30.56	44.10 ± 16.70	
SATrans-Net [33]	Acc.	68.06	47.92	73.96	53.47	61.11	45.14	73.61	71.53	63.19	62.00 ± 10.34	0.0039
	κ	57.41	30.56	65.28	37.96	48.15	26.85	64.81	62.04	50.93	49.30 ± 13.80	
SSM-DAN (source-only)	Acc.	68.06	51.04	86.46	61.11	64.24	39.58	72.92	73.61	62.15	64.35 ± 12.77	0.0039
	κ	57.41	34.72	81.94	48.15	52.31	19.44	63.89	64.81	49.54	52.50 ± 17.00	
Unsupervised domain adaptation												
CLUDA [34]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
RMETNet [35]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
MSD-DDA [36]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
SSM-DAN (UDA only)	Acc.	79.17	47.57	92.71	73.61	57.64	59.72	82.29	82.64	73.96	72.15 ± 13.58	0.0039
	κ	72.22	30.09	90.28	64.81	43.52	46.30	76.39	76.85	65.28	62.90 ± 18.10	
Test-time adaptation												
T-TIME [37]	Acc.	67.01	53.47	78.47	64.58	68.75	49.65	79.51	69.44	65.28	66.24 ± 9.34	0.0742
	κ	56.02	37.96	71.30	52.78	58.33	32.87	72.69	59.26	53.70	55.00 ± 12.50	
BFT [38]	Acc.	64.24	44.44	76.74	65.62	61.81	51.39	73.61	70.49	59.38	63.08 ± 9.77	0.0039
	κ	52.31	25.93	68.98	54.17	49.07	35.19	64.81	60.65	45.83	50.80 ± 13.00	
SSM-DAN (TTA only)	Acc.	77.43	47.92	88.19	66.32	64.24	61.81	79.86	82.99	73.61	71.37 ± 11.78	0.0039
	κ	69.91	30.56	84.26	55.09	52.31	49.07	73.15	77.31	64.81	61.80 ± 15.70	
SSM-DAN	Acc.	**79.51**	**52.43**	**93.40**	**73.96**	**69.44**	**65.97**	**88.89**	**86.11**	**75.35**	$\mathbf{76.12 \pm 11.96}$	–
	κ	**72.69**	**36.57**	**91.20**	**65.28**	**59.26**	**54.63**	**85.19**	**81.48**	**67.13**	$\mathbf{68.20 \pm 15.90}$	

4.2. BCI Competition IV-2b

Table 4 reports the completed binary left- versus right-hand results for the nine BCI IV-2b target subjects, whose recordings contain only three bipolar EEG derivations. Results use sessions 4–5 for the primary chronological evaluation; the standard deviation measures between-subject dispersion in this single LOSO run.

SSM-DAN achieved $84.09 \pm 10.42\%$ accuracy and 68.20 ± 20.80 Cohen's κ, compared with $80.54 \pm 10.05\%$ for its source-only variant and $83.97 \pm 10.71\%$ for its variant without test-time adaptation. SSM-DAN was 0.12 percentage points higher than the variant without test-time adaptation on average, but the paired difference was not significant ($p = 1.0000$). It significantly outperformed the source-only variant ($p = 0.0156$), and IM-TTA improved three of the nine target folds. SSM-DAN significantly exceeded ATCNet ($79.38 \pm 11.40\%$; $\kappa = 58.80 \pm 22.80$; $p = 0.0078$), which it outperformed on eight targets and tied on the ninth, and EEG Conformer ($78.03 \pm 12.02\%$; $\kappa = 56.10 \pm 24.00$; $p = 0.0156$), which it outperformed on eight of the nine targets. It also exceeded the matched TCFormer ($81.18 \pm 9.37\%$; $\kappa = 62.40 \pm 18.70$), EEGEncoder ($81.02 \pm 11.76\%$; $\kappa = 62.00 \pm 23.50$), and SATrans-Net ($81.76 \pm 8.78\%$; $\kappa = 63.50 \pm 17.60$) baselines on average, on six, six, and five of the nine targets, respectively, although none of these differences was significant ($p = 0.0820$, 0.0547, and 0.2109). The UDA baselines are compared with SSM-DAN (UDA only; $83.97 \pm 10.71\%$), which significantly exceeded RMETNet ($79.10 \pm 13.39\%$; $\kappa = 58.20 \pm 26.80$; $p = 0.0117$) and outperformed it on eight of the nine targets. Among the test-time adaptation baselines, BFT reached $81.18 \pm 12.12\%$ ($\kappa = 62.40 \pm 24.20$), 2.17 percentage points below its matched reference, SSM-DAN (TTA only; $83.35 \pm 10.50\%$; $\kappa = 66.70 \pm 21.00$); the reference was higher on eight targets and tied on the ninth ($p = 0.0078$).

4.3. Zhou2016

Table 5 uses source sessions 1–2 for training, source session 3 for validation, target sessions 1–2 for UDA, and target session 3 for primary scoring. With four folds, the smallest attainable two-sided Wilcoxon p-value is 0.125.

SSM-DAN reached $84.33 \pm 0.75\%$ accuracy and 76.50 ± 1.12 Cohen's κ, the highest mean and the lowest variability across targets. It exceeded the matched TCFormer baseline ($73.50 \pm 6.22\%$; $\kappa = 60.25 \pm 9.34$) and its source-only variant ($75.17 \pm 3.51\%$; $\kappa = 62.75 \pm 5.26$) on all four targets ($p = 0.1250$, the smallest attainable value with four folds). It also exceeded ATCNet

Table 4: Per-subject LOSO accuracy (%) and Cohen's κ multiplied by 100 on BCI IV-2b (two classes). p: Wilcoxon signed-rank test of accuracy against SSM-DAN for source-only methods and SSM-DAN variants, against SSM-DAN (UDA only) for unsupervised domain adaptation methods, and against SSM-DAN (TTA only) for test-time adaptation methods (Section 3.5). –: not yet available.

Method	Metric	Target subject									Mean ± SD	p
		S1	S2	S3	S4	S5	S6	S7	S8	S9		
Source-only training												
Conformer [6]	Acc.	75.31	62.86	55.62	92.50	94.69	74.06	82.81	82.50	81.88	78.03 ± 12.02	0.0156
	κ	50.63	25.71	11.25	85.00	89.38	48.12	65.62	65.00	63.75	56.10 ± 24.00	
ATCNet [7]	Acc.	76.25	65.71	57.19	90.00	96.88	77.50	84.06	84.69	82.19	79.38 ± 11.40	0.0078
	κ	52.50	31.43	14.38	80.00	93.75	55.00	68.12	69.38	64.38	58.80 ± 22.80	
TCFormer [8]	Acc.	78.44	69.64	64.38	90.62	95.00	75.94	86.87	85.31	84.38	81.18 ± 9.37	0.0820
	κ	56.88	39.29	28.75	81.25	90.00	51.88	73.75	70.63	68.75	62.40 ± 18.70	
EEGEncoder [32]	Acc.	73.44	65.71	59.06	92.50	97.19	85.00	85.94	86.56	83.75	81.02 ± 11.76	0.0547
	κ	46.88	31.43	18.12	85.00	94.38	70.00	71.88	73.12	67.50	62.00 ± 23.50	
SATrans-Net [33]	Acc.	77.50	69.29	69.69	90.62	97.50	79.69	87.50	84.38	79.69	81.76 ± 8.78	0.2109
	κ	55.00	38.57	39.38	81.25	95.00	59.38	75.00	68.75	59.38	63.50 ± 17.60	
SSM-DAN (source-only)	Acc.	77.81	68.57	63.13	89.38	96.56	73.44	84.69	86.25	85.00	80.54 ± 10.05	0.0156
	κ	55.62	37.14	26.25	78.75	93.12	46.88	69.38	72.50	70.00	61.10 ± 20.10	
Unsupervised domain adaptation												
CLUDA [34]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
RMETNet [35]	Acc.	73.12	57.86	55.00	91.56	94.06	82.81	85.62	88.75	83.13	79.10 ± 13.39	0.0117
	κ	46.25	15.71	10.00	83.13	88.12	65.62	71.25	77.50	66.25	58.20 ± 26.80	
MSD-DDA [36]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
SSM-DAN (UDA only)	Acc.	78.44	70.71	63.44	90.31	97.81	83.75	89.69	95.63	85.94	83.97 ± 10.71	1.0000
	κ	56.88	41.43	26.88	80.62	95.63	67.50	79.37	91.25	71.88	67.90 ± 21.40	
Test-time adaptation												
T-TIME [37]	Acc.	–	–	–	–	–	–	–	–	–	–	–
	κ	–	–	–	–	–	–	–	–	–	–	
BFT [38]	Acc.	75.00	65.00	59.69	93.44	94.69	79.06	88.12	94.37	81.25	81.18 ± 12.12	0.0078
	κ	50.00	30.00	19.37	86.88	89.38	58.13	76.25	88.75	62.50	62.40 ± 24.20	
SSM-DAN (TTA only)	Acc.	77.81	68.57	66.25	93.75	96.25	81.88	89.69	94.69	81.25	83.35 ± 10.50	0.2031
	κ	55.62	37.14	32.50	87.50	92.50	63.75	79.37	89.38	62.50	66.70 ± 21.00	
SSM-DAN	Acc.	77.81	69.29	65.94	90.00	97.19	84.69	91.56	95.31	85.00	**84.09 ± 10.42**	–
	κ	55.62	38.57	31.88	80.00	94.38	69.38	83.13	90.62	70.00	**68.20 ± 20.80**	

Table 5: Per-subject LOSO accuracy (%) and Cohen's κ multiplied by 100 on target session 3 of Zhou2016 (three classes) under the clean chronological protocol. Bold: best value in each column. p: Wilcoxon signed-rank test of accuracy against SSM-DAN for source-only methods and SSM-DAN variants, against SSM-DAN (UDA only) for unsupervised domain adaptation methods, and against SSM-DAN (TTA only) for test-time adaptation methods (Section 3.5). –: not yet available.

Method	Metric	Target subject				Mean ± SD	p
		S1	S2	S3	S4		
Source-only training							
Conformer [6]	Acc.	–	–	–	–	–	–
	κ	–	–	–	–	–	
ATCNet [7]	Acc.	72.00	82.67	76.67	74.67	76.50 ± 3.93	0.1250
	κ	58.00	74.00	65.00	62.00	64.80 ± 5.90	
TCFormer [8]	Acc.	68.00	78.67	66.67	80.67	73.50 ± 6.22	0.1250
	κ	52.00	68.00	50.00	71.00	60.25 ± 9.34	
EEGEncoder [32]	Acc.	62.00	84.00	74.67	76.67	74.33 ± 7.92	0.2500
	κ	43.00	76.00	62.00	65.00	61.50 ± 11.90	
SATrans-Net [33]	Acc.	81.33	80.00	67.33	70.00	74.67 ± 6.09	0.1250
	κ	72.00	70.00	51.00	55.00	62.00 ± 9.14	
SSM-DAN (source-only)	Acc.	72.00	78.67	71.33	78.67	75.17 ± 3.51	0.1250
	κ	58.00	68.00	57.00	68.00	62.75 ± 5.26	
Unsupervised domain adaptation							
CLUDA [34]	Acc.	–	–	–	–	–	–
	κ	–	–	–	–	–	
RMETNet [35]	Acc.	–	–	–	–	–	–
	κ	–	–	–	–	–	
MSD-DDA [36]	Acc.	–	–	–	–	–	–
	κ	–	–	–	–	–	
SSM-DAN (UDA only)	Acc.	79.33	84.67	**87.33**	78.00	82.33 ± 3.82	0.6250
	κ	69.00	77.00	**81.00**	67.00	73.50 ± 5.70	
Test-time adaptation							
T-TIME [37]	Acc.	–	–	–	–	–	–
	κ	–	–	–	–	–	
BFT [38]	Acc.	**84.00**	**86.67**	76.00	84.67	82.83 ± 4.07	0.8750
	κ	**76.00**	**80.00**	64.00	77.00	74.30 ± 6.10	
SSM-DAN (TTA only)	Acc.	79.33	84.00	83.33	83.33	82.50 ± 1.85	0.2500
	κ	69.00	76.00	75.00	75.00	73.75 ± 2.77	
SSM-DAN	Acc.	**84.00**	83.33	84.67	**85.33**	$\mathbf{84.33 \pm 0.75}$	–
	κ	**76.00**	75.00	77.00	**78.00**	$\mathbf{76.50 \pm 1.12}$	

($76.50 \pm 3.93\%$; $\kappa = 64.80 \pm 5.90$) on all four targets ($p = 0.1250$), SATransNet ($74.67 \pm 6.09\%$; $\kappa = 62.00 \pm 9.14$) on all four targets ($p = 0.1250$), and EEGEncoder ($74.33 \pm 7.92\%$; $\kappa = 61.50 \pm 11.90$) on three of the four targets ($p = 0.2500$). SSM-DAN (UDA only), which omits IM-TTA while retaining Stage 1 adaptation and AutoDIAL target-statistic re-estimation, reached $82.33 \pm 3.82\%$ ($\kappa = 73.50 \pm 5.70$; $p = 0.6250$) and is the matched reference of the UDA baselines; SSM-DAN (TTA only; $82.50 \pm 1.85\%$; $\kappa = 73.75 \pm 2.77$) is the matched reference of the test-time adaptation baselines. BFT reached $82.83 \pm 4.07\%$ ($\kappa = 74.30 \pm 6.10$), 0.33 percentage points above this reference on average (higher on three of the four targets; $p = 0.8750$), and obtained the highest accuracy on target 2 (86.67%). As an auxiliary diagnostic only, pooling all three target sessions gave $83.23 \pm 2.89\%$ ($\kappa = 74.85 \pm 4.34$) for SSM-DAN and $75.91 \pm 4.65\%$ ($\kappa = 63.90 \pm 7.00$) for TCFormer.

4.4. Ablation study

Table 6 follows the primary chronological protocol. The variant without domain adaptation removes every Stage 1 domain-adaptation component (AutoDIAL, target information maximization, and the adversarial and both MMD losses) while retaining Riemannian alignment and IM-TTA; all variants otherwise use the same folds and checkpoint rule.

Table 6: Ablation results under the primary chronological protocol: mean ± standard deviation of accuracy (%) and Cohen's κ (×100) across target subjects. BCI IV-2a is scored on session 2, BCI IV-2b on sessions 4–5, and Zhou2016 on session 3.

Dataset	Metric	SSM-DAN	Without Riemannian alignment	Without domain adaptation	Without test-time adaptation
BCI IV-2a	Acc.	$\mathbf{76.12 \pm 11.96}$	70.29 ± 11.60	71.37 ± 11.78	72.15 ± 13.58
	κ	$\mathbf{68.20 \pm 15.90}$	60.40 ± 15.50	61.80 ± 15.70	62.90 ± 18.10
BCI IV-2b	Acc.	$\mathbf{84.09 \pm 10.42}$	83.58 ± 10.46	83.35 ± 10.50	83.97 ± 10.71
	κ	$\mathbf{68.20 \pm 20.80}$	67.20 ± 20.90	66.70 ± 21.00	67.90 ± 21.40
Zhou2016	Acc.	$\mathbf{84.33 \pm 0.75}$	84.00 ± 3.16	82.50 ± 1.85	82.33 ± 3.82
	κ	$\mathbf{76.50 \pm 1.12}$	76.00 ± 4.70	73.75 ± 2.77	73.50 ± 5.70

On BCI IV-2a, removing Riemannian alignment, Stage 1 domain adaptation, or test-time adaptation reduced accuracy from 76.12% to 70.29%

($p = 0.0391$), 71.37% ($p = 0.0039$), and 72.15% ($p = 0.0039$), respectively. On BCI IV-2b, SSM-DAN (84.09%) was slightly above the variants without test-time adaptation (83.97%; $p = 1.0000$), without Riemannian alignment (83.58%; $p = 0.2578$), and without domain adaptation (83.35%; $p = 0.2031$). On Zhou2016, SSM-DAN (84.33%) was slightly above the variants without Riemannian alignment (84.00%), without domain adaptation (82.50%), and without test-time adaptation (82.33%); with only four folds, none of these differences can be significant (the smallest attainable p is 0.125; $p = 0.2500$ without domain adaptation and 0.6250 without test-time adaptation). Thus, SSM-DAN was best on all three datasets, although its BCI IV-2b variants were statistically indistinguishable.

4.5. Computational efficiency

Table 7 reports trainable parameters, Stage 1 training time, and CPU response time. The source-only variant is smaller because it omits the AutoDIAL layer, aligner, and domain discriminator; RA and IM-TTA do not change parameter count.

On Zhou2016, the source-only variant required 6.29 min per fold and 13.36 ms per CPU trial, compared with 11.42 min and 14.65 ms without test-time adaptation, 11.66 min and 18.88 ms without Riemannian alignment, 6.46 min and 18.41 ms without domain adaptation, and 11.70 min and 19.14 ms for SSM-DAN.

Table 7: Computational cost: trainable parameters, mean Stage 1 training time per fold, and single-trial CPU response time. UDA only: SSM-DAN without test-time adaptation; TTA only: SSM-DAN without domain adaptation; w/o RA: without Riemannian alignment.

Method	BCI IV-2a			BCI IV-2b			Zhou2016		
	Params (K)	Train (min)	CPU (ms)	Params (K)	Train (min)	CPU (ms)	Params (K)	Train (min)	CPU (ms)
Source-only training									
Conformer	789.6	11.14	11.45	759.1	4.93	9.34	–	–	–
ATCNet	113.7	48.58	13.33	112.8	60.19	11.35	113.3	8.09	11.91
TCFormer	127.2	38.37	14.96	123.4	31.04	11.17	125.6	5.89	15.97
EEGEncoder	197.2	13.49	26.51	196.2	14.98	26.88	196.8	3.64	25.68
SATrans-Net	155.6	10.19	12.78	153.7	4.99	11.10	155.0	1.32	11.38
SSM-DAN (source-only)	98.4	47.01	25.08	94.6	36.10	13.68	96.8	6.29	13.36
Unsupervised domain adaptation									
CLUDA	–	–	–	–	–	–	–	–	–
RMETNet	–	–	–	159.1	21.96	4.90	–	–	–
MSD-DDA	–	–	–	–	–	–	–	–	–
SSM-DAN (UDA only)	132.0	58.45	18.44	128.2	48.08	13.35	130.4	11.42	14.65
Test-time adaptation									
T-TIME	9.5	5.85	7.71	–	–	–	–	–	–
BFT	40.6	1.47	3.08	39.9	0.96	2.61	51.0	0.36	2.84
SSM-DAN (TTA only)	131.9	40.73	22.12	128.1	33.21	14.25	130.3	6.46	18.41
SSM-DAN w/o RA	132.0	58.41	21.63	128.2	47.95	13.33	130.4	11.66	18.88
SSM-DAN	**132.0**	**59.82**	**22.20**	**128.2**	**54.21**	**17.40**	**130.4**	**11.70**	**19.14**

5. Discussion

The results show that adaptation is dataset dependent. SSM-DAN was best on all three datasets, although on BCI IV-2b its margin over the variant without test-time adaptation was only 0.12 percentage points and all variants were statistically indistinguishable. On Zhou2016, SSM-DAN also had the lowest variability across targets, but its margins over the variants without Riemannian alignment, test-time adaptation, or Stage 1 domain adaptation were small and cannot be significant with four folds. Filtering to 8–30 Hz and excluding the first 0.5 s focus covariance estimation on MI-related activity rather than broadband drift and cue transients; repeated-seed experiments are still required to establish the stability of these differences.

Architecturally, multi-scale and depthwise convolutions preserve local spectral–spatial evidence, while alternating selective scans and the TCN model complementary long- and short-range dynamics. The alternating scans exploit complete offline trials and are non-causal; online use would require forward-only processing and device-specific latency validation.

The method is transductive because IM-TTA accesses unlabeled evaluation-session signals before scoring. Comparisons must therefore provide baselines with identical target access. Other limitations include the three-channel BCI IV-2b setting, the small Zhou2016 cohort, and the absence of repeated-seed and fully matched baseline results. Participant sex is reported only in aggregate or not at all for the public datasets, and the cohorts are too small for sex- or gender-stratified analysis; potential sex-related differences in decoding performance were therefore not assessed and limit the generalizability of the findings.

6. Conclusion

We presented SSM-DAN, a cross-subject MI-EEG framework combining multi-scale spatial–temporal convolution, alternating selective state-space blocks, local–global TCN fusion, hierarchical UDA, and constrained IM-TTA. Its chronological LOSO protocol excludes target labels from preprocessing, training, adaptation, and checkpoint selection. SSM-DAN obtained $76.12 \pm 11.96\%$ session-2 accuracy on BCI IV-2a and $84.09 \pm 10.42\%$ sessions 4–5 accuracy on BCI IV-2b, with corresponding Cohen's κ scores of 68.20 ± 15.90 and 68.20 ± 20.80 on the $\times 100$ scale. On Zhou2016, SSM-DAN obtained $84.33 \pm 0.75\%$ target-session-3 accuracy and 76.50 ± 1.12 Cohen's κ.

Matched baselines and repeated-seed experiments remain necessary before drawing broader conclusions.

CRediT authorship contribution statement

[Author name]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All EEG data analyzed in this study are publicly available. BCI Competition IV datasets 2a and 2b [29] and the Zhou2016 dataset [30] were accessed through the MOABB library [27]; no new data were collected.

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